Sturdevant's

ART and SCIENCE of OPERATIVE DENTISTRY

Seventh Edition

André V. Ritter | Lee W. Boushell | Ricardo Walter

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This book is dedicated to the art and science of Operative Dentistry. Our focus with this edition is to provide students, teachers, and practicing colleagues with a comprehensive and evidence-based resource for Operative Dentistry and related disciplines.

We also dedicate the seventh edition to the editors and contributors of the previous editions. Much of their work can still be found in this edition.

Finally, we dedicate this book to Drs. Clifford Sturdevant, Roger Barton, and John Brauer, who were the editors for The Art and Science of Operative Dentistry, First Edition, 1968. We hope that they would be proud to see how far their legacy has extended.

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Foreword

Dr. Clifford Sturdevant had a brass plaque on his desk that read "If it's almost right it's wrong!" This commitment to excellence also was the mantra upon which his classic textbook, *The Art and Science of Operative Dentistry,* was first written and published in 1968. This textbook has been the basis for training dental students in the fine art and clinical science of Operative Dentistry for 50 years. In light of this significant landmark, which coincides with the publication of this new Seventh Edition, we believe it is important to present the evolution of the various editions of the textbook from a historical perspective.

The First Edition (Sturdevant, Barton, Brauer, 1968) was meant "to present the significant aspects of Operative Dentistry and the research findings in the basic and clinical sciences that have immediate application" in the field of Operative Dentistry. It is important to note that Dean Brauer pointed out in his preface that beyond having the knowledge and skills needed to perform a procedure, the practitioner must also have high moral and ethical standards, essential and priceless ingredients. Since the First Edition, this textbook series has always attempted to present artistic and scientific elements of Operative Dentistry in the context of ethical standards for patient care.

It is also worth noting that the First Edition was printed and bound in "landscape" format so that it could more easily be used as a manual in the preclinical laboratory and would always remain open to the desired page. The handmade 5X models used to illustrate the various steps in cavity preparation were created by two dental students enrolled at The University of North Carolina at Chapel Hill School of Dentistry during the writing of the First Edition. Illustrations of these models have continued to be used in later editions, and the models themselves have served as important teaching materials for decades.

Although the techniques, materials, armamentarium, and treatment options continue to evolve, many of the principles of Operative Dentistry described in the First Edition are still pertinent today. An understanding of these principles and the ability to meticulously apply them are critical to providing the outstanding dental treatment expected by our patients.

The Second Edition (Sturdevant, Barton, Sockwell, Strickland, 1985) expanded on many techniques (e.g., acid etching) using experience and published research that had occurred since publication of the First Edition. The basics of occlusion were emphasized and presented in a way that would be helpful to the dental student and practitioner. A chapter on treatment planning and sequencing of procedures, as well as a chapter providing a thorough treatise on the use of pins, was included. Information on silicate cement, self-curing acrylic resin, and the baked porcelain inlay was eliminated for obvious reasons. A chapter on endodontic therapy and the chapter on the "dental assistant" were no longer included. Chapters on (1) tooth-colored restorations and (2) additional conservative and esthetic treatments explained the changes and improvements that occurred in the areas of esthetic options available to patients. In the chapter on gold inlay/onlay restorations, increased emphasis was given to the gold onlay restorations for Class II cavity preparations.

The Third Edition (Sturdevant, Roberson, Heymann, J. Sturdevant, 1995) placed a new emphasis on cariology and the "medical model of disease" with regard to risk assessment and managing the high-risk caries patient. This important concept laid the foundation for what is still taught today with regard to identifying risk factors and defining a treatment plan based on caries risk assessment. The Third Edition also included new expanded chapters on infection control, diagnosis and treatment planning, and dental materials. In light of the growing interest in the area of esthetic dentistry, a variety of conservative esthetic treatments were introduced including vital bleaching, microabrasion and macroabrasion, etched porcelain veneers, and the novel all-porcelain bonded pontic. Additionally, an entirely new section on tooth-colored inlays and onlays was included that chronicled both lab-processed resin and ceramic restorations of this type and those fabricated chairside with CAD/CAM systems.

With the Fourth Edition of this text (Roberson, Heymann, Swift, 2002), Dr. Clifford Sturdevant's name was added to the book title to honor his contributions to the textbook series and the discipline of Operative Dentistry. In this edition, a particular emphasis was placed on bonded esthetic restorations. Consequently, an entirely new chapter was included on fundamental concepts of enamel and dentin adhesion. This chapter was intended to provide foundational information critical to the long-term success of all types of bonded restorations.

The Fifth Edition (Roberson, Heymann, Swift, 2006) continued with the renewed emphasis on the importance of adhesively bonded restorations and focused on scientific considerations for attaining optimal success, particularly with posterior composites. Concepts such as the "C Factor" and keys to reducing polymerization effects were emphasized along with factors involved in reducing microleakage and recurrent decay.

The Sixth Edition (Heymann, Swift, Ritter, 2013) represented a transition from a large printed edition, as in the past, to a smaller, streamlined printed version that focused on concepts and techniques immediately essential for learning contemporary Operative Dentistry. The same amount of information was included, but many chapters such as those addressing biomaterials, infection control, pain control, bonded splints and bridges, direct gold restorations, and instruments and equipment were available for the first time in a supplemental online format.

With this new Seventh Edition of *Sturdevant's Art and Science of Operative Dentistry*, fundamental concepts and principles of contemporary Operative Dentistry are maintained and enhanced,

but vital new areas of content also have been incorporated. Diagnosis, classification, and management of dental caries have been significantly updated in light of the latest clinical and epidemiological research. Similarly, content on adhesive dentistry and composite resins has been updated as a result of the evolving science in these fields.

An entirely *new* chapter on light curing and its important role in the clinical success of resin composite restorations has been added. Moreover, a *new* scientifically based chapter details the important elements of color and shade matching and systematically reviews how the dental clinician is better able to understand the many co-variables involved in color assessment. It also reviews how best to improve shade matching of esthetic restorations to tooth structure.

In an attempt to better optimize restorative treatment outcomes involving periodontal challenges, a *new* chapter has been included that addresses these principles. Periodontology Applied to Operative Dentistry chronicles the various clinical considerations involving conditions such as inadequate crown length, lack of root coverage, and other vexing problems requiring interdisciplinary treatment to optimize success.

Finally, the Seventh Edition of this text addresses the ever-evolving area of digital dentistry with a *new* chapter, Digital Dentistry in Operative Dentistry. This chapter reviews the various technologies involved in scanning and image capture for both treatment planning and restorative applications. Additionally, the authors review various types of digital restorative systems for both chairside and modem-linked laboratory-based fabrication of restorations. In recognition of the rapid movement to digital dentistry, this chapter is a vital addition to a textbook whose tradition has been always to reflect the latest technologies and research findings in contemporary Operative Dentistry.

Since its inception 50 years ago, the Sturdevant text has been a dynamic document, with content that has included innovative information on the latest materials and techniques. Over this time period, numerous internationally recognized experts have addressed many specific topics as authors and co-authors of various chapters. It also should be pointed out that with all editions of the textbook, the authors of the various chapters are themselves actively involved in teaching students preclinical and clinical Operative Dentistry. Moreover, they are "wet-fingered dentists" who also practice Operative Dentistry for their individual patients.

In summary, for 50 years *Sturdevant's Art and Science of Operative Dentistry* has been a major resource guiding educators in the teaching of contemporary Operative Dentistry. Each edition of this text has striven to incorporate the latest technologies and science based on the available literature and supporting research. The Seventh Edition is a superb addition to this tradition, which will most assuredly uphold the standard of publication excellence that has been the hallmark of the Sturdevant textbooks for half a century.

Harald O. Heymann, DDS, MEd
Kenneth N. May, Jr., DDS

Preface

Since the publication of the First Edition in 1968, The University of North Carolina's *The Art and Science of Operative Dentistry* has been considered a major Operative Dentistry textbook in many countries, and it has been translated in several languages. The widespread use of this textbook in dental education is a testimony to both its success and its value for dental students and dental educators alike.

With the Seventh Edition we attempted to elevate the level of excellence this textbook series is known for. All relevant content from the previous editions is still here (from cariology and treatment planning to biomaterials and clinical techniques for amalgam and composite restorations). However, most chapters were significantly revised to reflect current scientific and clinical evidence, and several chapters were virtually rewritten by new contributors who are more engaged in the specific content areas. The chapter on biomaterials, in addition to being significantly revised, appears again in print in the Seventh Edition, making it easier for the reader to access the information while reading the print version of the textbook. Additionally, many chapters were condensed into more streamlined and concise single chapters (for example, three of the "composite chapters" from the Sixth Edition are now concisely presented in a single "composite chapter" in this new edition; a similar approach was used for the "amalgam chapters").

In addition, four new and relevant chapters were added (Light Curing of Restorative Materials, Color and Shade Matching in Operative Dentistry, Periodontology Applied to Operative Dentistry, and Digital Dentistry in Operative Dentistry) to bring the textbook in line with disciplines that ever more interface with Operative Dentistry, emphasizing the increased role of an interdisciplinary approach to modern Operative Dentistry. Each of these new chapters is authored by recognized authorities in the respective topics, and considerably broaden the scope of the Seventh Edition.

The new edition also features an Expert Consult website that includes a full online version of the print text, as well as five additional online-only chapters and technique videos.

Expanding on a significant layout facelift that started with the Sixth Edition, the Seventh Edition offers an increased number of color images, line drawings that were further revised and improved for increased text comprehension, and a reorganization of chapter sequence. Furthermore, redundant and outdated information has been deleted. All these updates enhance the user experience and make the Seventh Edition an even user-friendlier textbook for the wide range of readers—students, teachers, and practitioners/colleagues.



Dr. Clifford Sturdevant.

Many hours of diligent work have been invested so as to offer you the best possible Operative Dentistry textbook at this point in time. We have sought to honor the long-standing tradition of *The Art and Science of Operative Dentistry* textbook series and to bring you updated, clinically relevant and evidence-based information. To publish this edition on the year we commemorate the 50th anniversary of the publication of the First Edition is a milestone for Operative Dentistry in general and for The University of North Carolina at Chapel Hill's Department of Operative Dentistry in particular. We are honored to have had the opportunity to work on and present the Seventh Edition.

The Editors

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- Our spouses and families for their love, understanding, and support during this revision.
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